

# Find&Define: Automatic Extraction of Definitions from Text

## Progress Report

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The automatic extraction of definitions from text using machine learning techniques is a growing field of research. Typically this was always done in a limited, structured and well-defined way using pattern matching. Recently, more advanced techniques such as neural networks are being used to solve this problem. The goal of this study is to use a variety of different types of neural networks to extract textual definitions from educational papers, books and other formal writings.

Additional Key Words and Phrases: Definition Extraction, Natural Language Processing

## 1 INTRODUCTION

Definitions are the main way humans are able to communicate and transfer knowledge about the meanings of concepts. In order to help generalise and share these definitions, dictionaries were created. However, as new concepts are invented, or old ones are updated or get outdated, these definitions in dictionaries also have to be updated. Linguists typically go through this process several times. Although this means that the definitions would have a certain standard, it also means that dictionaries are rarely ever up to date. This is especially the case when it comes to scientific domains which evolve and update very frequently. This is exactly where the automatic extraction of textual definitions can help a lot. This process of extracting term-definition pairs from text corpora is known as Definition Extraction (DE).

Definition extraction is not a new concept, for well over a decade, definition extraction has been a studied topic in NLP research, but it has mostly been limited to well-defined, structured, and narrow conditions since this was more often than not, done using structural and pattern matching. [Muresan and Klavans 2002; Saggion 2004; Storrer and Wellinghoff 2006; Walter and Pinkal 2006]

Natural language, however, is in reality, complicated, and complicated data necessitates both complex solutions and data that reflects this reality. The main complications are that firstly, term-definition pairs appear in different sentences and secondly, that there is a lack of explicit definitions in some sentences. Not having explicit definitions means not having the words which show definition such as; is, means, is defined as, etc. Nowadays, researchers have shifted to extracting definitions using machine learning techniques or a hybrid between machine learning and pattern matching. [Fahmi and Bouma 2006; Miliaraki and Androutsopoulos 2004]

There are several areas in which definition extraction can be useful. Apart from the previously mentioned construction of dictionaries, definitions can also help in the building of knowledge graphs. By getting to know if two terms are related or not, you may add new edges and nodes to the graph, which help increase the connectivity and relevance of the graph. This can in return help in question answering machines, which benefit from better knowledge graphs apart from just the definitions.

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## 1.1 Aims and Objectives

This project aims to extract textual definitions from educational papers, books and other formal writings. Splitting the problem into three smaller problems, the aims are:

- (1) Sentence Classification: Classify whether a sentence contains a definition or not.
- (2) Sequence Labeling: Each token is labelled with BIO tags.
- (3) Relation Classification: Label the relations between each tag in a tag sequence.

There are different approaches which could be taken in order to fulfil these aims. Therefore a good evaluation plan should be developed, which would help in deciding which model would be the most appropriate to use, taking into consideration any additional layers or pre-processing steps which could be added. In order to achieve this, the following objectives are defined:

- (1) Choose the dataset/s and understand the structure and define what pre-processing would be needed.
- (2) Experiment with different techniques to identify the possible approaches which could be taken.
- (3) Decide on whether or not to work with fine-tuning a pre-built system or creating a new one from scratch.
- (4) Experiment with different parameters and addition of any layers.
- (5) Define an in-depth evaluation plan in order to benchmark results and be able to compare.

## 2 LITERATURE REVIEW

Definition extraction is a niche topic in natural language processing, one of which is definition extraction. A corporation that explores this topic is SemEval. The SemEval-2020, which is a series of international natural language processing (NLP) research workshops and competitions, included a competition for definition extraction, namely Task 6 [Spala et al. 2020], sometimes called by DeftEval for short. This included three sub-tasks on the DEFT corpus. These sub-tasks aimed to provide an idea of the workflow needed to finally achieve definition extraction.

The first sub-task focuses on sentence classification. Sub-task 1 is used to classify whether or not a sentence contains a definition. The corresponding sentence is marked as '1' if the definition is included. As stated in the introduction, this can be done by using textual cues such as 'is', 'means', etc. However, a definition can also be present without such words. These should not be ignored.

The second subtask is a sequence labelling task. Sub-task 2 is used to label each word in a sentence, including punctuation, with BIO tags according to the corpus' tag specification. BIO tags are essentially a way of classifying the 'type' of a given token by using B (Begin), I (Interior), O (Outside) and sometimes E (End). Having a B-prefix before a tag signifies that the tag is at the start of a chunk, while an I-prefix denotes that the tag is within a chunk. When a

tag is followed by another tag of the same type with no O tokens between them, the B-tag is used. An O-tag shows that a token does not belong to any entity or chunk.

The third subtask is a relation classification task. Sub-task 3 uses the tag sequence labels to label the relations between each tag, also based on the corpus' relation specifications.

This served as a good starting to see what technologies are currently used, which achieved better results, and so which to ignore and which to look further into.

## 2.1 DeftEval Competition

[Spala et al. 2020]

In their paper Jeawak et al. 2020 focus just on the first sub-task of the SemEval-2020 in their paper. They discuss two different approaches. A single BERT model based on all available training data; fine-tuning a separate BERT model for each of the 7 domains, each time relying only on the training data available for that domain, and a hybrid approach combining both of these methods. The first strategy has the advantage of being able to use all of the training data. However, they believe that this strategy will have difficulty capturing the various definition styles used in different domains. The second strategy avoids the confusion generated for the classifier with different definition styles, but it does mean that each domain has a finite amount of training data. Finally, the hybrid strategy requires fine-tuning a domain-specific BERT model in two steps: first on all training data, then on the target domain in a second fine-tuning step. Jeawak et al. experimented with the BERT-base approach, mostly using the BertForSequenceClassification PyTorch huggingface implementation. Essentially, this approach requires adding a classification layer on top of a pre-trained BERT model and fine-tuning the BERT model while training the classification layer. They also experiment with various methodologies such as BERT-all, BERT-target, BERT-double, BERT-name. Aside from that, the team also experimented with LSTMs LSTMpaper, employing contextualised word vectors from BERT as input and testing numerous methodologies including LSTM-base, LSTM-BERT, LSTM-BERT-ft.

Looking at the results, fine-tuning BERT beats LSTM-based techniques in terms of performance. When the various fine-tuning tactics were compared, it was discovered that specifying the domain name as an additional token in BERT-name failed to outperform the usual fine-tuning strategy. The usual technique outperformed domain-specific fine-tuning on average. However, the double fine-tuning technique produced the best overall performance, having been trained on both general and domain-specific contexts. Despite the fact that double fine-tuning produced the best results, the team still chose the BERT-all standard method since it was the most efficient and produced nearly equivalent results.

Davletov et al. employ a Transformer-based model trained concurrently for all three tasks in their system. On top of the Transformer output, they add a linear layer with a dropout for each task. This enables the system to use knowledge about sentence types, entities, and relationships between them while training. This helps in the improvement of the outcomes for Subtasks 1 and 3. However,

for Sub-task 2, a single-task model performed better. The primary contribution of this paper is a detailed analysis of the performance of multi-task systems for definition extraction, classification, and named entity recognition. The results demonstrate that while this method of multi-task training may be useful for sequence classification, it should be reconsidered for sequence labelling. Davletov et al. proposes a model for multi-task learning models that predicts an input example class, a tag sequence of entity labels, and semantic relations between entities all at the same time. To address this, they treat relation extraction as a sequence labelling problem (similar to how named entity recognition is usually solved). For single-task models, however, the researchers fine-tuned a Roberta large model [Liu et al. 2019b] for the first sub-task and the BERT-large-uncased model for sub-task 2, with each token taking the BERT embedding from the first sub-token and passing it via a dropout layer followed by a linear layer.

Avram et al. investigate multiple pre-trained language models and display their findings for the first and second subtasks, respectively. As has been demonstrated on other corpora (Peters et al., 2019), they discovered that fine-tuning the weights of the pre-trained language models on the DEFT corpus improves performance over freezing them. Furthermore, they look into a RoBERTa model [Liu et al. 2019a] that is within a multi-task architecture that jointly learns the outputs of the second and third subtasks. However, only its performance on the second subtask was reported, since they underperformed on the third. For the first subtask, their initial approach was to use a convolutional neural network followed by a bidirectional long short-term memory model; however, they decided to use a pre-trained model that would be trained better than they would have otherwise obtained while also being more convenient and fast because it was already available. The researchers treat the second subtask as a sequence labelling problem, and hence build their method on prior work such as part-of-speech tagging [Yang et al. 2019], named entity recognition [Ma and Hovy 2016][Dumitrescu and Avram 2019], metaphor detection [Wu et al. 2018], emphasis selection [Shirani et al. 2019], key phrase extraction [Alzaidy et al. 2019], fragment-level propaganda detection [Da San Martino et al. 2019], complex word identification [Gooding and Kochmar 2019], language identification [Mave et al. 2018], or cyber-attacks detection in server logs [Ghimes et al. 2018]. Avram et al. did not take into account the fact that they were given the set of correct tags from the second subtask (which greatly aids in relation classification) for the third and final subtask, and thus their approaches obtained poor results on both development and test datasets, so the third and final subtask was omitted and was not discussed in the rest of the paper.

The researchers observed an imbalance in the different categories when they first processed the data. For example, there are 11,090 sentences in the first subtask that do not have a definition and 5,569 that do. To deal with this, they equalise the classes by doubling the number of phrases with definitions, resulting in a new total of 11,138 positive samples. Similarly, the second task has extremely unbalanced classes, with the Definition tag having a class count of 93,204 and the Referential-Term tag having a count of 256. In this scenario, Avram et al. over-sampled each sentence containing an

under-represented tag by a factor inversely proportionate to their quantity to equalise the classes (by using this method, the number of over-represented tags increases as well, but by a much smaller factor). The researchers also chose to pre-process the dataset further, removing any artefacts that would interfere with the language model's representation of the sentences. During the fine-tuning process and when freezing the language model weights, for example, they replace URLs and equations with two specific tokens. They also remove artefacts from text formatting, which are frequently caused by expressions, such as 'size 12' and spaces before punctuation. They also remove characters that were not recognised by the language models and could break the tag-sub token matching process, such as the "o" character or Greek letters, in the second subtask. They also opted to swap out the accented characters with their non-accented counterparts.

BERT, XLNet, RoBERTa, SciBERT, and ALBERT were among the five pre-trained language models employed by the researchers, each of which was further tested by freezing and fine-tuning its weights. Fine-tuning the weights of the language models provides a significant performance gain, as seen in the above Table of results. Furthermore, the results on the development dataset reveal that, for both subtasks 1 and 2, a fine-tuned RoBERTa is the best-performing language model among all others. The model was found to be slightly biased towards predicting positive labels in the first subtask, resulting in more false positives for the input sentences, which is to be expected given the balancing technique used.

Avram et al. also mentions that they plan to examine the multi-task situation using various language models in the future. They also believe that combining a larger annotated dataset with large versions of the pre-trained language models could boost the performance of definition extraction significantly. Furthermore, they claim that by employing class weights instead of oversampling for the first subtask, some issues in the results can be mitigated.

Zhang and Ren uses BERT to address the multi-domain problems presented in SemEval 2020 task 6. They attained a best performance of 79.71 per cent in F1 in subtask 1 using BERT, BiLSTM, and attentiveness. For subtask 2, BERT, BiLSTM, and CRF were used for sequence labelling, producing a Macro-averaged F1 of 40.73 percent. They use PyTorch-Transformers in the system, which is a library of great pre-trained Natural Language Processing (NLP) models. They use PyTorch transformers to invoke the BERT module. BertModel, BertConfig, and BertTokenizer are three modules. Also, use an uncased version of Base BERT. AdamW was the chosen module optimizer, which is implemented in the PyTorch-Transformers module<sup>1</sup>. When it comes to CRF, they use PyTorch-CRF<sup>2</sup> to call CRF modules. During the training phase, the ReduceLROnPlateau is applied to dynamically reduce the learning rate.

Subtask 1 accepts sentences separated by a line break as input, but also the special marks '[CLS]' and '[SEP]' added at the beginning and end of the sentence, respectively. This data is subsequently processed by a BERT layer to obtain word representations, and then

by a BiLSTM and an attention layer to obtain semantic information. The key reason they choose an LSTM is that it may serve as a gate mechanism, controlling the amount of prior information retained by each LSTM unit and remembering the current input information, retaining only the key info and discarding the rest. However, since LSTM does not take backward knowledge into account, this caused a new issue. This was addressed by employing a BiLSTM model rather than a standard LSTM model, which mixes forward and backward LSTM. Their approach can now capture the context of information at any point in the sentence. The importance of this was emphasised by Zhou et al. In terms of the attention layer, different weights are assigned to indicate that different words have varying degrees of relevance. This was expressed using the following formulas (also by Zhou et al.):

(1)

$$M = \tanh(HT * \omega)$$

(2)

$$\alpha = \text{softmax}(M * \mu T)$$

(3)

$$\gamma = H * \alpha T$$

In which H refers to the output of BiLSTM,  $\alpha$  refers to the value of attention,  $\gamma$  refers to the output of the attention layer,  $\mu$  and  $\omega$  are hyper-parameters.

Subtask 2 uses a similar architecture to subtask 1, but they also make use of a CRF layer [Huang et al. 2015], which may leverage sentence-level tag information to select the label with the highest score for each word. Although subtask2 can be solved without CRF, CRF will provide some constraints to assure the validity of the final prediction results, as well as reduce invalid results.

## 2.2 Other Research

The work by Roig Mirapeix et al. is not part of the DeftEval Competition, but it uses the DEFT corpus, which was developed for the competition, as one of its corpora. The primary goal of this project was to investigate how far it is possible to teach a universal definition extraction system from canonical definitions, as well as to understand the key differences that now exist in conventional assessment testbeds. Aside from the DEFT corpus, they also utilize the Word-Class Lattices (WCL) dataset from Navigli et al. [Navigli et al. 2010], which is an annotated set of definitions and non-definitions with syntactic patterns comparable to those of Wikipedia definition sentences.

With regards to the choice of model, they train a supervised model using a Support Vector Machine (SVM) due to its efficiency and effectiveness in handling a vast amount of linguistic features. The model employs an RBF kernel and a variety of characteristics. The primary one is based on tagged sentences with n-grams ranging from 1 to 3, i.e. each word has a chunk tag, a Part-of-speech (PoS) tag, and a word separated by an underscore.

For each training set, the model computes the 5 most common definition verbs, i.e. in the main part of the definition, the 20 most common hypernyms, the 10 most common composition of chunk and PoS tags, the 6 most common chunk tags, the 10 most common structures of chunk and PoS tags combined, the 10 most common

<sup>1</sup><https://pypi.org/project/pytorch-transformers/>

<sup>2</sup><https://pypi.org/project/pytorch-crf/0.7.2/>

structures of chunk tags, and the maximum length of definitions. As a result of this, the following additional characteristics are obtained:

- VERB: The number of common verbs in the sentence.
- HYP: The number of popular hypernyms in the sentence.
- CT-Ch, CT-Ch&PoS: The number of occurrences divided by the total number of tags in the phrase for each of the six most common chunk tags and the ten most common combinations of chunk and PoS.
- STR-Ch, STR-Ch&PoS: A binary variable showing whether each of the ten most common structures (chunk and combinations of chunk and PoS, respectively) is present in the phrase.
- LEN: The length of the sentence divided by the definition's maximum length.

Nonetheless, the authors of this work, Roig Mirapeix et al., demonstrated that a simple SVM model trained on a dataset with canonical definitions utilising linguistic features can deliver good performance while also helping us in better understanding the problem. This approach was also tested on a corpus with diverse domains, which provided us with insights into the qualitative differences between definitions in each situation.

They also mention that it would be interesting to expand the research to corpora with other characteristics and languages in the future. A model with accurate performance across corpora would allow the automatic construction of dictionaries from general or specialised domains, as well as greater understanding of specific topics.

### 2.3 Datasets Available

A perfect algorithm is useless without a good dataset therefore multiple datasets were considered before deciding on using the DEFT Corpus. Five datasets were considered in total. The 'MIMIC-III' [Johnson et al. 2016] is a freely accessible critical care database comprising information relating to patients admitted to critical care units at a large tertiary care hospital, such as symptoms, procedure codes and diagnostic codes. Although the dataset contained textual definitions of what the diagnostics were, this dataset was omitted since it would have required a substantial amount of pre-processing, tagging and formatting.

'BioInfer' [Pyysalo et al. 2007] is a similar corpus focusing on information extraction for the biomedical domain. Although this dataset showed potential it had some limitations. Having about 1,100 sentences, it is on the smaller side. The BioInfer Corpus is also found in one in.xml file which is also not very well structured. Therefore this corpus was also put aside.

The next dataset considered was the 'Genia' [Collier et al. 1999] corpus which was built with bio-text mining. It is a large corpus, offering different file formats, which were all well structured. However since this data was structured for Part-of-Speech annotation, it is not very applicable for definition extraction.

Another corpus that was considered as the one used in the paper "Automatic Definition Extraction Using Evolutionary Algorithms" by [Borg et al. 2010]. Unfortunately, this is not publicly available anymore. (European LT4eL project <sup>3</sup>)

<sup>3</sup><https://cordis.europa.eu/project/id/027391>

The final dataset considered was the [Spala et al. 2019] DEFT Corpus which was constructed specifically for definition extraction and is the largest annotated, semi-structured corpus for this field. This corpus consists of sentences from two different sources. The first source is contract filing from publicly available from the US Securities and Exchange Commission EDGAR (SEC) database. The second part is sentences taken from open-source textbooks found on <https://cnx.org>, covering a number of different topics in biology, history, physics, psychology, economics, sociology, and government. SemEval 2020 Task 6 (DeftEval), a definition extraction competition utilised this dataset, from which participants managed to achieve very good results. This competition also serves as a good starting point to get an idea of what techniques worked best.

After looking at these datasets, it was decided that the DEFT Corpus would be used. Although the biomedical corpora were more interesting in the sense of having an actual application, the DEFT corpus offers a concrete structure specifically designed for the problem at hand.

## 3 PROPOSED SOLUTION

Given these possible solutions, it was decided that, building a language model from scratch, did not make much sense. When looking at research done by others, the results achieved by existing models, even in other fields other than that of definition extraction, is significantly greater than custom made models, and so, building a worse or 'bad' model seems redundant.

Therefore, the question arises as to which model should be used. The best option seems to be some variation of BERT, such as RoBERTa. However, apart from this, additional layers can also be added to the model as well.

Although it is too early to decide exactly what the model to be used will consist of, a set of tests should be carried out to figure out which would be the most suitable model which should include the following models; BERT, RoBERTa, BiLSTM, Support Vector Machines, LSTMs, attention layer, linear layers, and segmented training of the dataset.

## 4 EVALUATION PLAN

Proposed evaluation metrics for the DeftEval 2020 (SemEval 2020 - Task 6) competition <sup>4</sup>

- (1) 1. Subtask 1: Sentence Classification: We will report P/R/F1 for the positive and negative classes. The official score will be based on the F1 for the positive class.
- (2) 2. Subtask 2: Sequence labelling: We will report P/R/F1 for each evaluated class, as well as macro- and micro-averaged F1 for the evaluated classes. The official score will be based on the macro-averaged F1 of the evaluated classes. Evaluated classes include Term, Alias-Term, Referential-Term, Definition, Referential-Definition, and Qualifier.
- (3) 3. Subtask 3: Relation extraction: We will report P/R/F1 for each evaluated relation, as well as macro- and micro-averaged F1 for the evaluated relations. The official score will be based on the macro-averaged F1 of the evaluated relations. The

<sup>4</sup>[https://competitions.codalab.org/competitions/22759#learn\\_the\\_details-evaluation](https://competitions.codalab.org/competitions/22759#learn_the_details-evaluation)

evaluated relations include Direct-defines, Indirect-defines, Refers-to, AKA, and Qualifies.

These metrics could originally be run locally by using the evaluation code available on the DEFT Github repo<sup>5</sup>, however, this is only available for those who enter the competition, which is now closed. The test set contains data from the same distribution of data as the train and dev sets. This would open the possibility of comparing the obtained results with those achieved by participants of the SemEval 2020.

## 5 CONCLUSION

This project goes over the research carried out prior to attempting to implement Automatic Extraction of Definitions from Text. The deft corpus was chosen because it is specifically built with definition extraction in mind, contains all necessary train, dev and test sets and has a lot of documentation explaining any annotations. There are several possible approaches that could be taken for the implementation, some testing will be done in order to decide exactly which model variation would yield the best results considering the processing power available. F1 score will be used to decide the correctness and accuracy of the results, having the ability to compare the results with those of similar research in the same domain.

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